

Real Time Research Project

Digital Shield Rail Defense

An AI-Driven Platform for Combating Human Trafficking on Indian Railways through Real-Time
Detection and Intelligent Response

Abstract:

Human trafficking on Indian Railways is a critical public safety challenge, exploiting the scale and anonymity of a network carrying over 23 million passengers daily. RPF officers currently depend on physical observation alone, making detection reactive and inconsistent. Digital Shield Rail Defense addresses this by simultaneously analysing two real-time data streams — live CCTV footage and the IRCTC ticketing database. A Machine Learning risk scoring engine flags anomalous booking patterns such as minors travelling with unrelated adults or last-minute bulk one-way tickets, while a Computer Vision model detects behavioural indicators of coercion on camera feeds. When both streams flag the same group, a corroborated alert is pushed instantly to the nearest RPF officer's mobile application.

Python powers the backend for ML inference, NLP intake, and statistical analysis. PostgreSQL manages cases and alerts, Redis handles real-time queuing, and Neo4j enables suspect network link analysis. The frontend delivers a React Native officer app and a React.js command dashboard connected via a FastAPI REST layer. An NLP pipeline converts free-text citizen reports into structured case data with automatic face redaction. The system follows six pillars — Detection, Reporting, Intelligence, Passenger Protection, Officer Enablement, and Ethics Governance — with Explainable AI ensuring every alert states the specific reason it was generated. The project demonstrates how machine learning, computer vision, graph analytics, and responsible AI principles can be combined into a deployable anti-trafficking intelligence platform.

TOOLS:

Frontend	React Native (RPF Officer App — iOS & Android), React.js (Command Dashboard), HTML, CSS, JavaScript
Backend	Python (FastAPI) — Alert Engine, Risk Scoring API, NLP Inference, Media Processing
Runtime & API	Node.js, REST API, WebSockets (Real-Time Alert Push)
Database	PostgreSQL (Cases, Alerts, Users), Redis (Alert Queue), Neo4j (Suspect Link Graph)
ML / AI / NLP	scikit-learn, XGBoost (Risk Scoring), YOLOv8 (Computer Vision), spaCy, IndicBERT (NLP Intake), NumPy, pandas
Data Structures	Queues (Alert Processing), Graphs (Network Link Analysis), Arrays (Case & Portfolio Tracking)
Development Tools	VS Code, GitHub, Docker, Figma, Postman

** Subject to revision based on project requirements and infrastructure availability.*

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